

transmitted from the Wi-Fi module can be accessed on a remote device like a smartphone or laptop using ThingSpeak's website.

ESP8266 Wi-Fi module has two GPIO pins that can be used for connecting to the Internet through an MCU using specific firmware. However, these are not used in this project and it works on the default firmware.

ESP8266 Wi-Fi module is used here to connect to your Wi-Fi modem and upload temperature data to ThingSpeak. You can view that data from a smartphone or laptop in the form of a plotted-line graph.

In this project, an LCD is also used as a user interface device to see circuit configuration status and temperature data. Data lines (D0 through D7) of the LCD are connected to port B of IC4.

Software

espIOT code for PIC16F887 MCU is written in C language. It is compiled using MPLAB IDE version 8.91 and HI-TECH C compiler version 9.83.0.10920. The generated hex code is burnt into the MCU chip using MPLAB ICD 2 programmer board.

Construction and testing

An actual-size, single-side PCB layout of the Wi-Fi temperature logger is shown in Fig. 6, and its component layout in Fig. 7. When the circuit is switched on, the program initialises the LCD and flashes welcome message 'ESP Temp Logger' on the LCD.

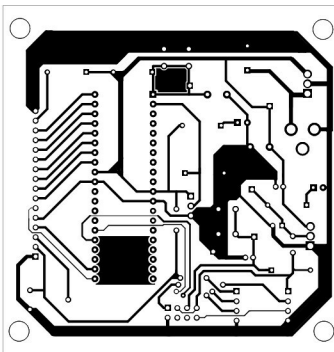


Fig. 6: Actual-size PCB layout of the Wi-Fi temperature-logging system

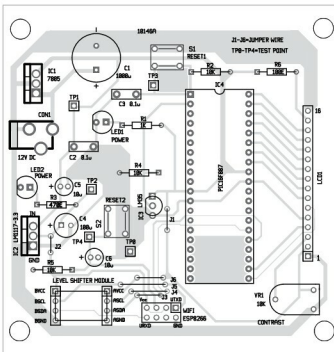


Fig. 7: Component layout of the PCB

If there is any display problem on the LCD, check voltages at various points as listed in the test point table. If the LCD is showing a proper message, go to www.thingspeak.com. First, sign in to www.thingspeak.com.

com and then create your own channel by creating a field as shown in Fig. 8.

After saving the channel, you will receive channel ID and two API keys—a write API key (it is to be used in our C code) and a read API key (it is used to check the data from website on your smartphone or PC). Make a note of these keys as you will need these later.

Before using the system, ensure that ESP8266 module baud rate is changed to 9600 as most of these have 115200. Also, a Wi-Fi modem with an Internet connection should be kept near your circuit board.

Next, make the following changes in the espIOT.C code:

1. Replace eRocks14 with your own Wi-Fi name
2. Replace eRocks2014 with your own Wi-Fi password
3. Replace API key with your own API key received

ESP8266 works on basic AT commands similar to the ones used in any GSM module interfacing project. It can be easily checked and tested from HyperTerminal of Windows. The link to AT commands is www.pridopia.co.uk/pi-doc/ESP8266ATCommandsSet.pdf

In the latest version of ESP8266 firmware, baud rate setting command AT + CIOBAUD is no longer

supported. You can replace this command by AT + IPR. The detail of this command can be found on www.esp8266.com/viewtopic.php?f=13&t=718

When you open your chan-